AWS Certified AI Practitioner

AIF-C01 — Quiz Set (30 Questions)

Thirty original, blueprint-weighted questions across all five domains, tiered by difficulty (Recall / Application / Analysis) and including multiple-response items. Every answer explains the causal **why** and why each distractor is wrong. Use the elimination method: read the last sentence first, classify the task, cut categorically-wrong options, break the tie on the constraint, then check the distractor pairs.

Domain	Weight	Questions
D3 — Applications of Foundation Models	28%	8
D2 — Fundamentals of Generative AI	24%	7
D1 — Fundamentals of AI & ML	20%	6
D4 — Responsible AI	14%	5
D5 — Security, Compliance & Governance	14%	4

Answer key with full explanations begins after the question set.

Questions

Q1 · D3 · Application

1. A startup wants to add generative chat to its app. It has no ML engineers, does not want to manage servers or model hosting, and wants to call several model providers through one API. Which service fits best?

- **A.** Amazon SageMaker Studio
- **B.** Amazon Bedrock
- **C.** Amazon SageMaker JumpStart
- **D.** Amazon EC2 with an open-source model

Q2 · D3 · Analysis

2. A bank's assistant must answer from policy PDFs that are revised every month, and answers must reflect the latest version. The team wants the least development effort to add this knowledge. Which approach?

- **A.** Fine-tune the foundation model monthly on the new PDFs
- **B.** Increase the model's temperature
- **C.** Use a Bedrock Knowledge Base (RAG) over the PDFs
- **D.** Add all PDFs into every prompt

Q3 · D3 · Application

3. A team already uses Amazon Bedrock and must stop the assistant from discussing a banned topic and from emitting customers' Social Security numbers. Which capability addresses both?

- **A.** SageMaker Clarify
- **B.** Bedrock Guardrails
- **C.** Amazon Macie
- **D.** Provisioned throughput

Q4 · D3 · Recall

4. Which scenario is the strongest fit for SageMaker JumpStart rather than Amazon Bedrock?

- **A.** A team with no ML skills wanting a serverless API
- **B.** Deploying an open-source foundation model inside the company's own VPC
- **C.** Filtering harmful content from an LLM
- **D.** Detecting PII in an S3 bucket

Q5 · D3 · Application

5. A marketing team wants product blurbs in a specific brand voice with the least operational effort, and can supply a few good examples. Which prompt technique?

- **A.** Zero-shot prompting
- **B.** Few-shot prompting
- **C.** Fine-tuning
- **D.** Continued pre-training

Q6 · D3 · Application

6. An application must return the scanned key-value fields from thousands of uploaded invoices. Which AWS service is purpose-built for this?

- **A.** Amazon Comprehend
- **B.** Amazon Rekognition

- **C.** Amazon Textract
- **D.** Amazon Kendra

Q7 · D3 · Analysis

7. A support tool must, in one request, look up an order via an internal API, check a returns policy document, and draft a reply — coordinating multiple steps. Which Bedrock capability is designed for this?

- **A.** Bedrock Guardrails
- **B.** Bedrock Agents
- **C.** On-demand pricing
- **D.** Titan Embeddings

Q8 · D3 · Application

8. A workload scores a large batch of records overnight with no latency requirement. Which SageMaker inference option is most appropriate and cost-effective?

- **A.** Real-time endpoint
- **B.** Asynchronous inference
- **C.** Batch transform
- **D.** Serverless inference

Q9 · D2 · Recall

9. What best defines a foundation model?

- **A.** A small model trained for one narrow task
- **B.** A large, general-purpose model pre-trained on broad data that can be adapted to many tasks
- **C.** A rules engine for model deployment
- **D.** A database of embeddings

Q10 · D2 · Application

10. When comparing foundation models on Bedrock, a team asks how much text can be sent and returned in a single request. Which property answers this?

- **A.** Temperature
- **B.** Context window
- **C.** Top-p
- **D.** Model parameter count

Q11 · D2 · Analysis

11. A team is evaluating an abstractive summarization model against reference summaries. Which metric pair is appropriate?

- **A.** Accuracy and F1
- **B.** RMSE and R^2
- **C.** ROUGE and BERTScore
- **D.** Precision and recall of pixels

Q12 · D2 · Recall

12. In a GAN, what is the discriminator's job?

- **A.** Generate new synthetic samples
- **B.** Classify inputs as real or fake
- **C.** Store embeddings

D. Reduce dimensionality

Q13 · D2 · Application

13. A model confidently outputs a plausible but entirely invented statistic. What is this called, and which is a primary mitigation?

A. Overfitting; add regularization
B. Hallucination; ground responses with RAG
C. Drift; retrain weekly
D. Bias; run Clarify

Q14 · D2 · Recall

14. Which are Amazon's own (first-party) foundation models on Bedrock?

A. The Titan family
B. Anthropic Claude
C. Meta Llama
D. Mistral

Q15 · D2 · Analysis · Select TWO

15. A company will customize (fine-tune) an FM on proprietary data. Which TWO are the primary trade-offs? (Select TWO)

A. Higher cost
B. Reduced accuracy
C. Higher implementation complexity
D. Guaranteed elimination of hallucination
E. Lower latency

Q16 · D1 · Application

16. A model shows 98% accuracy on training data but 71% on held-out test data. What is happening?

A. Underfitting
B. Overfitting
C. Data leakage into test set
D. Too few epochs

Q17 · D1 · Recall

17. Which of the following is a hyperparameter?

A. A learned weight
B. The learning rate
C. A predicted label
D. A model bias term learned in training

Q18 · D1 · Application

18. Which task is an example of unsupervised learning?

A. Predicting house prices from features
B. Classifying emails as spam or not
C. Grouping customers into segments without labels
D. Forecasting next month's sales

Q19 · D1 · Application

19. A team wants to raise model accuracy toward an acceptance threshold by giving the model more passes over the training data. Which lever?

- **A.** Increase the number of epochs
- **B.** Increase temperature
- **C.** Decrease the batch size to 1
- **D.** Reduce the feature count

Q20 · D1 · Recall

20. Which SageMaker capability best matches “label a large image dataset using humans plus ML assistance”?

- **A.** SageMaker Feature Store
- **B.** SageMaker Ground Truth
- **C.** SageMaker Model Monitor
- **D.** SageMaker Canvas

Q21 · D1 · Application

21. A business analyst with no coding skills wants to build a predictive model through a visual interface. Which SageMaker tool?

- **A.** SageMaker Studio
- **B.** SageMaker Canvas
- **C.** SageMaker Autopilot
- **D.** SageMaker Pipelines

Q22 · D4 · Application

22. A regulated healthcare model must be transparent and its predictions explainable to auditors. Which service provides bias detection and explainability?

- **A.** Amazon Macie
- **B.** SageMaker Clarify
- **C.** Bedrock Guardrails
- **D.** Amazon Inspector

Q23 · D4 · Analysis

23. Which correctly distinguishes Bedrock Guardrails from SageMaker Clarify?

- **A.** Guardrails explains model predictions; Clarify blocks topics
- **B.** Guardrails filters generative content at runtime; Clarify analyzes bias/explainability at training/eval
- **C.** Both are network-security tools
- **D.** Both are pricing models

Q24 · D4 · Application

24. An image-generation model's training data is skewed toward one demographic, producing biased outputs. Which mitigation acts at the source?

- **A.** Raise temperature
- **B.** Data augmentation to rebalance under-represented classes
- **C.** Switch to batch inference
- **D.** Add more epochs

Q25 · D4 · Application

25. A document-processing pipeline should send only low-confidence predictions to a human for review. Which service implements this?

- **A.** Amazon A2I (Augmented AI)
- **B.** Bedrock Guardrails
- **C.** SageMaker Model Cards
- **D.** Amazon Comprehend

Q26 · D4 · Recall

26. What is the purpose of SageMaker Model Cards?

- **A.** Serve real-time predictions
- **B.** Record intended uses, risk ratings, training details, and evaluation results for governance
- **C.** Detect PII in S3
- **D.** Tune hyperparameters automatically

Q27 · D5 · Application

27. A security team needs a record of every Amazon Bedrock API call — which principal called which action and when — to investigate unauthorized access attempts. Which service?

- **A.** Amazon CloudWatch
- **B.** AWS CloudTrail
- **C.** Amazon Inspector
- **D.** AWS Trusted Advisor

Q28 · D5 · Application

28. A company must find and classify PII (SSNs, credit-card numbers) sitting in its S3 buckets. Which service?

- **A.** Amazon Inspector
- **B.** Amazon Macie
- **C.** AWS Config
- **D.** AWS Artifact

Q29 · D5 · Analysis

29. Using the generative-AI security scoping model, which option places the MOST security responsibility on the customer?

- **A.** Using a third-party SaaS app with embedded generative AI
- **B.** Building an app on a third-party FM API
- **C.** Fine-tuning a third-party FM with company data
- **D.** Building and training a model from scratch on company-owned data

Q30 · D5 · Application

30. A company wants to secure who is allowed to invoke Amazon Bedrock, following least privilege. Which service is central?

- **A.** AWS IAM
- **B.** Amazon Macie
- **C.** Amazon CloudWatch
- **D.** AWS X-Ray

Answer key & explanations

Q1 — Correct: B [D3 · Application]

Why. The decisive constraints are “no ML engineers,” “no server/host management,” and “one API for several providers” — the definition of Bedrock, a serverless multi-provider FM service.
Why the others are wrong. A — Studio is the full custom-build IDE — maximum management, the opposite of the constraint. **C** — JumpStart still hands you an endpoint to operate and is single-model-centric. **D** — Self-hosting on EC2 is the most operational overhead of all.

Q2 — Correct: C [D3 · Analysis]

Why. “Latest version” + “changes monthly” + “least development effort to add knowledge” = RAG. A managed Knowledge Base retrieves current documents at inference with no retraining.
Why the others are wrong. A — Monthly fine-tuning is high, recurring effort and still lags the newest data. **B** — Temperature changes randomness, not knowledge access. **D** — Stuffing every PDF blows the context window and doesn't scale.

Q3 — Correct: B [D3 · Application]

Why. Denied-topic policies and PII redaction are Guardrails features applied to inputs/outputs at runtime.
Why the others are wrong. A — Clarify measures ML bias/explainability, not runtime content filtering. **C** — Macie finds sensitive data in S3; it doesn't filter chatbot output. **D** — Provisioned throughput is a pricing/capacity option.

Q4 — Correct: B [D3 · Recall]

Why. “Open-source model” + “inside our VPC” signals owning the endpoint — JumpStart provides pre-built models you deploy into your environment.
Why the others are wrong. A — Serverless/no-skills is the Bedrock signal. **C** — That's Guardrails. **D** — That's Macie.

Q5 — Correct: B [D3 · Application]

Why. “A few examples” + “least operational effort” + style guidance = few-shot: examples in the prompt steer voice without any training.
Why the others are wrong. A — Zero-shot gives no examples, so voice is less controlled. **C** — Fine-tuning needs many labeled pairs and real training effort. **D** — Continued pre-training is the heaviest option, for whole new domains.

Q6 — Correct: C [D3 · Application]

Why. Extracting text and key-value/table data from scanned documents and forms is exactly Textract (OCR).
Why the others are wrong. A — Comprehend analyzes the meaning of text, it doesn't OCR documents. **B** — Rekognition analyzes images/video for objects and faces. **D** — Kendra is enterprise search over indexed content.

Q7 — Correct: B [D3 · Analysis]

Why. Multi-step orchestration that calls APIs and knowledge sources and chains results is the purpose of Bedrock Agents.
Why the others are wrong. A — Guardrails filters content; it doesn't orchestrate steps. **C** — Pricing model, unrelated. **D** — Embeddings support retrieval, not task orchestration.

Q8 — Correct: C [D3 · Application]

Why. “Large batch, overnight, no latency need” = offline processing = Batch Transform.
Why the others are wrong. A — A persistent real-time endpoint is for live, low-latency traffic. **B** — Async suits large payloads needing a result within ~an hour, still request-driven. **D** — Serverless is for intermittent real-time calls without managing infra.

Q9 — Correct: B [D2 · Recall]

Why. Foundation models are large, broadly pre-trained, and adaptable via prompting/RAG/fine-tuning — the premise of Bedrock.
Why the others are wrong. **A** — That describes a narrow, task-specific (often fine-tuned) model. **C** — That is not a model at all. **D** — An embedding store is infrastructure, not a model.

Q10 — Correct: B [D2 · Application]

Why. The context window is the max tokens (input + output) per request — the capacity being asked about.
Why the others are wrong. **A** — Temperature controls randomness. **C** — Top-p controls sampling diversity. **D** — Parameter count relates to capability, not per-request text capacity.

Q11 — Correct: C [D2 · Analysis]

Why. Summarization is text generation: ROUGE (n-gram overlap) and BERTScore (semantic similarity) are the right family.
Why the others are wrong. **A** — Accuracy/F1 are classification metrics. **B** — RMSE/R^2 are regression metrics. **D** — Pixel metrics are for imaging, not text.

Q12 — Correct: B [D2 · Recall]

Why. The discriminator classifies real vs fake while the generator tries to fool it; the competition improves both.
Why the others are wrong. **A** — That's the generator. **C** — Not a GAN component role. **D** — Unrelated technique.

Q13 — Correct: B [D2 · Application]

Why. Confident fabrication is hallucination; grounding the model in retrieved real data (RAG) is a primary mitigation.
Why the others are wrong. **A** — Overfitting is a training-generalization problem, not fabrication at inference. **C** — Drift is changing input distributions over time. **D** — Bias/fairness is a different responsible-AI issue.

Q14 — Correct: A [D2 · Recall]

Why. Titan (Text, Embeddings, Image) is AWS's first-party FM family; the others are third-party providers on Bedrock.
Why the others are wrong. **B** — Claude is Anthropic's. **C** — Llama is Meta's. **D** — Mistral is a separate provider.

Q15 — Correct: A, C [D2 · Analysis]

Why. Customization buys task fit at the price of training/compute cost and engineering complexity.
Why the others are wrong. **B** — Accuracy generally improves — that's the goal. **D** — Fine-tuning reduces but does not guarantee removal of hallucination. **E** — Latency is not the customization trade-off in question.

Q16 — Correct: B [D1 · Application]

Why. High train / low test accuracy is the textbook signature of overfitting — the model memorized noise and fails to generalize.
Why the others are wrong. **A** — Underfitting is poor on BOTH sets. **C** — Leakage would inflate test scores, not depress them. **D** — Too few epochs usually underfits (low on both).

Q17 — Correct: B [D1 · Recall]

Why. Hyperparameters are set before training (learning rate, batch size, epochs). Weights/biases are learned during training.
Why the others are wrong. **A** — Weights are learned parameters. **C** — A prediction is an output. **D** — Bias terms are learned parameters, not hyperparameters.

Q18 — Correct: C [D1 · Application]

Why. Clustering unlabeled customers into segments is unsupervised — no labels, structure discovered from data.
Why the others are wrong. A — Regression (supervised). **B** — Classification (supervised). **D** — Forecasting from labeled history (supervised).

Q19 — Correct: A [D1 · Application]

Why. An epoch is one full pass through the training data; more epochs give more learning opportunities (up to overfitting).
Why the others are wrong. B — Temperature is an inference parameter, not training. **C** — Batch size changes gradient behavior, not directly "more passes." **D** — Fewer features can lose signal.

Q20 — Correct: B [D1 · Recall]

Why. Ground Truth is the labeling service combining human labelers with ML-assisted (active) labeling.
Why the others are wrong. A — Feature Store reuses engineered features. **C** — Model Monitor watches production drift. **D** — Canvas is no-code model building.

Q21 — Correct: B [D1 · Application]

Why. "No code / business analyst / visual" = Canvas.
Why the others are wrong. A — Studio is the full code IDE. **C** — Autopilot is AutoML with visibility (still more technical than Canvas). **D** — Pipelines orchestrate ML workflows.

Q22 — Correct: B [D4 · Application]

Why. Clarify detects bias and explains predictions (e.g., SHAP feature attribution) — the transparency/explainability control.
Why the others are wrong. A — Macie finds sensitive data in S3. **C** — Guardrails filters generative content, not explainability. **D** — Inspector scans infrastructure vulnerabilities.

Q23 — Correct: B [D4 · Analysis]

Why. Guardrails = runtime content safety on inputs/outputs; Clarify = statistical bias and explainability analysis. Different layers.
Why the others are wrong. A — Reversed roles. **C** — Neither is a network-security tool. **D** — Neither is a pricing model.

Q24 — Correct: B [D4 · Application]

Why. Bias enters through imbalanced data, so the source fix is augmenting/rebalancing under-represented classes.
Why the others are wrong. A — Temperature affects randomness, not fairness. **C** — Inference mode is irrelevant to bias. **D** — More epochs can entrench the existing bias.

Q25 — Correct: A [D4 · Application]

Why. A2I routes predictions below a confidence threshold to human reviewers — human-in-the-loop oversight.
Why the others are wrong. B — Guardrails auto-blocks content; it doesn't route to humans by confidence. **C** — Model Cards document models. **D** — Comprehend is an NLP service.

Q26 — Correct: B [D4 · Recall]

Why. Model Cards are immutable governance records of a model's intended uses, risks, and evaluation — audit/compliance documentation.
Why the others are wrong. A — That's an endpoint. **C** — That's Macie. **D** — That's Autopilot/HPO.

Q27 — Correct: B [D5 · Application]

Why. “Who called what action, when” = API audit logging = CloudTrail.
Why the others are wrong. **A** — CloudWatch tracks metrics/alarms (health), not who-did-what. **C** — Inspector scans for vulnerabilities. **D** — Trusted Advisor gives best-practice checks.

Q28 — Correct: B [D5 · Application]

Why. Discovering and classifying sensitive data in S3 is Macie's core function.
Why the others are wrong. **A** — Inspector scans infrastructure/code vulnerabilities (EC2/ECR). **C** — Config tracks resource configuration changes. **D** — Artifact provides compliance reports.

Q29 — Correct: D [D5 · Analysis]

Why. Security responsibility scales with how much of the stack you own; training from scratch is maximum ownership.
Why the others are wrong. **A** — Embedded SaaS is least customer responsibility. **B** — Building on an API shifts less than fine-tuning. **C** — Fine-tuning is more than API-use but less than full training.

Q30 — Correct: A [D5 · Application]

Why. Authorizing which principals can perform which actions, with least privilege, is IAM.
Why the others are wrong. **B** — Macie is data discovery. **C** — CloudWatch is monitoring. **D** — X-Ray is request tracing.